What will I Learn from this AI course?

Experience a cutting-edge AI curriculum designed by distinguished IIT Hyderabad faculty, backed by practical, hands-on learning.

Module 1: **Regression and Classification**

Python for AI

- Linear Regression
- Logistic Regression
- Naive Bayes classifier and SVMs
- Decision Trees and Random Forests

Lab Activities

- Implementing regression and classification
- Models Model evaluation and validation

Module 2: **Ensemble Methods, Clustering, Dimensionality Reduction**

Lecture Topics

- Ensemble Methods (e.g., Bagging and Boosting)
- Clustering Algorithms (e.g., K-Means, DBSCAN)
- Dimensionality Reduction (e.g., PCA)

Lab Activities

- Applying clustering and dimensionality reduction techniques
- Building ensemble models

Module 3: **Neural Networks and Deep Learning**

Lecture Topics

- Perceptrons, MLP
- Convolutional Neural Networks (CNNs)
- Recurrent Neural Networks (RNNs)

Lab Activities

- Pytorch Framework
- Implementing neural networks
- Transfer learning with pre-trained models

Module 4: **Computer Vision, speech and natural language processing**

Python for AI

- Computer Vision (Classification, Object Detection and Object Segmentation)
- Speech Processing (Problems and Methodologies)
- Natural Language Processing (NLP) (Problems and Methodologies)

Lab Activities

- Hands-on with image processing, speech and NLP tasks
- Building AI applications

Module 5: **Advanced Topics in Deep Learning**

Lecture Topics

- Deep learning for vision and NLP
- Encoder-Decoder Models, Transformers

Lab Activities

- Implementing and training advanced deep learning models

Module 6: **Generative AI and Agentic AI**

Lecture Topics

- VAE, GAN and diffusion models for vision
- Large language models - training, finetuning, prompt engineering
- Agentic
- AI Responsible AI

Lab Activities

- Exploring LLM , prompt engineering in LLMs

Advanced Projects

Students work on their selected projects under continuous mentoring and expert guidanceSample projects:

- E-commerce Recommendations
- Object Recognition Image
- classification and other vision problems
- Text classification and other NLP problems
- Spam Filtering with NLP
- Detecting Fake News
- Deep Learning and Generative AI

Tools covered